

FastAPI Application Startup Failure SOP

Objective

Restore FastAPI service startup when the server crashes during initialization due to syntax errors, dependency issues, or missing configuration.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T018
Issue	FastAPI app not starting
Category	Deployment
Service	FastAPI
Severity	High
Status	Resolved
Description	Server crashes on startup
Expected Resolution	Check dependency versions and syntax errors

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm pod, container, or process startup failure and capture restart count.
2. Review startup logs for syntax errors, import errors, missing environment variables, or dependency failures.
3. Validate Python version, package versions, and application entrypoint.
4. Check Uvicorn or Gunicorn command, module path, and application object name.
5. Validate database migrations, startup hooks, and external dependency calls.
6. Roll back recent dependency or configuration changes if needed.
7. Redeploy corrected application image or configuration.
8. Confirm health endpoint passes and restart count remains stable.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
App not starting	Import or syntax error	Fix code and redeploy
Module not found	Missing dependency	Update requirements and rebuild image
Startup hook fails	Dependency unavailable	Add timeout or fix dependency
CrashLoopBackOff	Config or runtime failure	Check logs and environment variables

Best Practices

1. Run startup smoke tests in CI/CD.
2. Pin critical dependency versions.
3. Keep startup hooks lightweight.
4. Expose a reliable health endpoint.

References

[FastAPI Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [API Gateway Logs](#) | [Backend Runbook](#)